

$$\begin{aligned} \text{HD} \rightarrow & \frac{1}{16\pi^2} \left(\frac{2}{27} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_{\mathbf{d}}^{\text{P}}] - \frac{5}{36} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{d}}^{\text{P}}] + \right. \\ & \frac{8}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{d}}^{\text{P}}] + \frac{2}{9} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_{\mathbf{e}}^{\text{P}}] - \frac{5}{12} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{e}}^{\text{P}}] + \\ & \frac{8}{45} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{e}}^{\text{P}}] + \left(-\mathbf{c}_{2\gamma} \mathbf{g}_2^2 \mathbf{c}_{\gamma^2} \overline{\mathbf{y}}_{\mathbf{e}}^{\text{Pr}} \mathbf{y}_{\mathbf{e}}^{\text{Pr}} + \frac{1}{36} \sum_{\mathbf{p}} \left(4 \mathbf{g}_1^4 + 9 \mathbf{g}_2^4 \mathbf{c}_{2\gamma}^2 \right) \right) \text{LF}_{3,0}[\mathbf{m}_{\mathbf{l}}^{\text{P}}] + \\ & \left(\mathbf{c}_{2\gamma} \mathbf{g}_2^2 \mathbf{c}_{\gamma^2} \overline{\mathbf{y}}_{\mathbf{e}}^{\text{Pr}} \mathbf{y}_{\mathbf{e}}^{\text{Pr}} - \frac{1}{24} \sum_{\mathbf{p}} \left(5 \mathbf{g}_1^4 + 6 \mathbf{g}_2^4 \mathbf{c}_{2\gamma}^2 \right) \right) \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{l}}^{\text{P}}] + \frac{4}{45} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{l}}^{\text{P}}] + \\ & \left(\frac{1}{27} \sum_{\mathbf{p}} \mathbf{g}_1^4 + \frac{3}{4} \mathbf{c}_{2\gamma} \mathbf{g}_2^2 \left(-4 \mathbf{c}_{\gamma^2} \overline{\mathbf{y}}_{\mathbf{d}}^{\text{Pr}} \mathbf{y}_{\mathbf{d}}^{\text{Pr}} + 4 \mathbf{s}_{\gamma^2} \overline{\mathbf{y}}_{\mathbf{u}}^{\text{Pr}} \mathbf{y}_{\mathbf{u}}^{\text{Pr}} + \sum_{\mathbf{p}} \mathbf{c}_{2\gamma} \mathbf{g}_2^2 \right) \right) \text{LF}_{3,0}[\mathbf{m}_{\mathbf{q}}^{\text{P}}] + \\ & \left(3 \mathbf{c}_{2\gamma} \mathbf{g}_2^2 \left(\mathbf{c}_{\gamma^2} \overline{\mathbf{y}}_{\mathbf{d}}^{\text{Pr}} \mathbf{y}_{\mathbf{d}}^{\text{Pr}} - \mathbf{s}_{\gamma^2} \overline{\mathbf{y}}_{\mathbf{u}}^{\text{Pr}} \mathbf{y}_{\mathbf{u}}^{\text{Pr}} \right) - \frac{1}{72} \sum_{\mathbf{p}} \left(5 \mathbf{g}_1^4 + 54 \mathbf{g}_2^4 \mathbf{c}_{2\gamma}^2 \right) \right) \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{q}}^{\text{P}}] + \\ & \frac{4}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{q}}^{\text{P}}] + \frac{8}{27} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_{\mathbf{u}}^{\text{P}}] - \frac{5}{9} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{u}}^{\text{P}}] + \frac{32}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{u}}^{\text{P}}] + \\ & \frac{1}{576} \left(\mathbf{g}_1^4 \left(73 + 9 \mathbf{c}_{4\gamma} \left(-2 + \mathbf{c}_{4\gamma} \right) - 36 \mathbf{s}_{2\gamma}^4 \right) + 18 \mathbf{g}_1^2 \mathbf{g}_2^2 \left(-3 + \mathbf{c}_{4\gamma} \left(2 + \mathbf{c}_{4\gamma} \right) - 4 \mathbf{s}_{2\gamma}^4 \right) + \right. \\ & \quad 9 \mathbf{g}_2^4 \left(\left(3 + \mathbf{c}_{4\gamma} \right)^2 - 4 \mathbf{s}_{2\gamma}^4 \right) \left. \right) \text{LF}_{3,0}[\mathbf{m}_{\mathbb{H}}] + \frac{1}{192} \left(-6 \mathbf{g}_1^2 \mathbf{g}_2^2 \left(-3 + \mathbf{c}_{4\gamma} \left(2 + \mathbf{c}_{4\gamma} \right) - 4 \mathbf{s}_{2\gamma}^4 \right) + \right. \\ & \quad 3 \mathbf{g}_2^4 \left(-\left(3 + \mathbf{c}_{4\gamma} \right)^2 + 4 \mathbf{s}_{2\gamma}^4 \right) + \mathbf{g}_1^4 \left(-43 - 3 \mathbf{c}_{4\gamma} \left(-2 + \mathbf{c}_{4\gamma} \right) + 12 \mathbf{s}_{2\gamma}^4 \right) \left. \right) \text{LF}_{4,-1}[\mathbf{m}_{\mathbb{H}}] + \\ & \frac{4}{45} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbb{H}}] + \frac{1}{9} \mathbf{g}_1^4 \text{LF}_{3,0}[\tilde{\mu}] + \frac{1}{6} \mathbf{g}_1^4 \text{LF}_{4,-1}[\tilde{\mu}] - \frac{8}{45} \mathbf{g}_1^4 \text{LF}_{5,-2}[\tilde{\mu}] + \\ & \frac{1}{8} \mathbf{g}_1^4 \text{LF}_{2,2,-1}[\mathbf{m}_1, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 + \frac{1}{4} \mathbf{g}_1^4 \mathbf{m}_1^2 \text{LF}_{2,2,0}[\mathbf{m}_1, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 + \\ & \frac{17}{8} \mathbf{g}_2^4 \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 + \frac{1}{4} \mathbf{g}_2^4 \mathbf{m}_2^2 \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 - \\ & 4 \mathbf{g}_2^4 \text{LF}_{3,2,-2}[\mathbf{m}_2, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 + 3 \mathbf{g}_2^4 \mathbf{m}_2^2 \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 + \\ & 2 \mathbf{g}_2^4 \text{LF}_{3,3,-3}[\mathbf{m}_2, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 - 2 \mathbf{g}_2^4 \mathbf{m}_2^2 \text{LF}_{3,3,-2}[\mathbf{m}_2, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 + \\ & 2 \mathbf{g}_2^4 \text{LF}_{4,2,-3}[\mathbf{m}_2, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 - 2 \mathbf{g}_2^4 \mathbf{m}_2^2 \text{LF}_{4,2,-2}[\mathbf{m}_2, \tilde{\mu}] \left(\mathbf{c}_{\gamma^2} - \mathbf{s}_{\gamma^2} \right)^2 + \\ & \frac{2}{3} \mathbf{g}_1^2 \left(\mathbf{c}_{\gamma} \overline{\mathbf{a}}_{\mathbf{d}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \overline{\mathbf{y}}_{\mathbf{d}}^{\text{Pr}} \right) \left(\mathbf{c}_{\gamma} \mathbf{a}_{\mathbf{d}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \mathbf{y}_{\mathbf{d}}^{\text{Pr}} \right) \text{LF}_{2,2,0}[\mathbf{m}_{\mathbf{d}}^{\text{r}}, \mathbf{m}_{\mathbf{q}}^{\text{P}}] - \\ & \frac{1}{3} \mathbf{g}_1^2 \left(\mathbf{c}_{\gamma} \overline{\mathbf{a}}_{\mathbf{d}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \overline{\mathbf{y}}_{\mathbf{d}}^{\text{Pr}} \right) \left(\mathbf{c}_{\gamma} \mathbf{a}_{\mathbf{d}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \mathbf{y}_{\mathbf{d}}^{\text{Pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_{\mathbf{d}}^{\text{r}}, \mathbf{m}_{\mathbf{q}}^{\text{P}}] + \\ & \frac{2}{3} \mathbf{g}_1^2 \left(\mathbf{c}_{\gamma} \overline{\mathbf{a}}_{\mathbf{e}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \overline{\mathbf{y}}_{\mathbf{e}}^{\text{Pr}} \right) \left(\mathbf{c}_{\gamma} \mathbf{a}_{\mathbf{e}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \mathbf{y}_{\mathbf{e}}^{\text{Pr}} \right) \text{LF}_{2,2,0}[\mathbf{m}_{\mathbf{e}}^{\text{r}}, \mathbf{m}_{\mathbf{l}}^{\text{P}}] - \\ & \frac{1}{3} \mathbf{g}_1^2 \left(\mathbf{c}_{\gamma} \overline{\mathbf{a}}_{\mathbf{e}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \overline{\mathbf{y}}_{\mathbf{e}}^{\text{Pr}} \right) \left(\mathbf{c}_{\gamma} \mathbf{a}_{\mathbf{e}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \mathbf{y}_{\mathbf{e}}^{\text{Pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_{\mathbf{e}}^{\text{r}}, \mathbf{m}_{\mathbf{l}}^{\text{P}}] - \\ & \frac{2}{3} \mathbf{g}_1^2 \left(\mathbf{c}_{\gamma} \overline{\mathbf{a}}_{\mathbf{e}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \overline{\mathbf{y}}_{\mathbf{e}}^{\text{Pr}} \right) \left(\mathbf{c}_{\gamma} \mathbf{a}_{\mathbf{e}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \mathbf{y}_{\mathbf{e}}^{\text{Pr}} \right) \text{LF}_{3,1,0}[\mathbf{m}_{\mathbf{l}}^{\text{P}}, \mathbf{m}_{\mathbf{e}}^{\text{r}}] - \\ & \frac{2}{3} \mathbf{g}_1^2 \left(\mathbf{c}_{\gamma} \overline{\mathbf{a}}_{\mathbf{e}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \overline{\mathbf{y}}_{\mathbf{e}}^{\text{Pr}} \right) \left(\mathbf{c}_{\gamma} \mathbf{a}_{\mathbf{e}}^{\text{Pr}} - \mathbf{s}_{\gamma} \tilde{\mu} \mathbf{y}_{\mathbf{e}}^{\text{Pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_{\mathbf{l}}^{\text{P}}, \mathbf{m}_{\mathbf{e}}^{\text{r}}] + \\ & \frac{1}{2} \left(3 \mathbf{g}_1^2$$